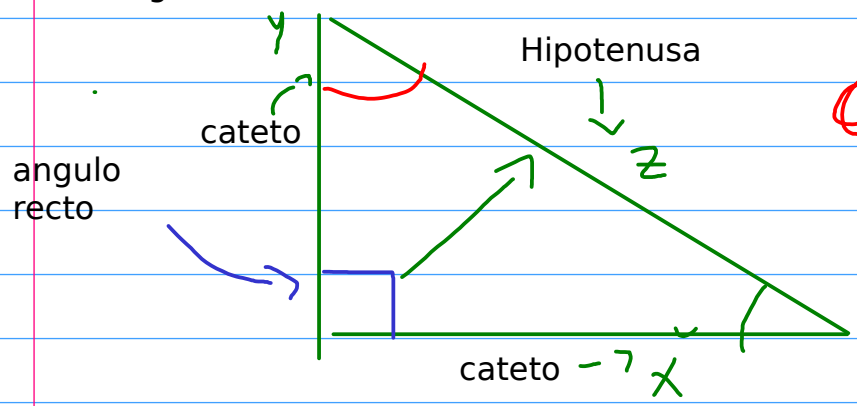
Pitagoras

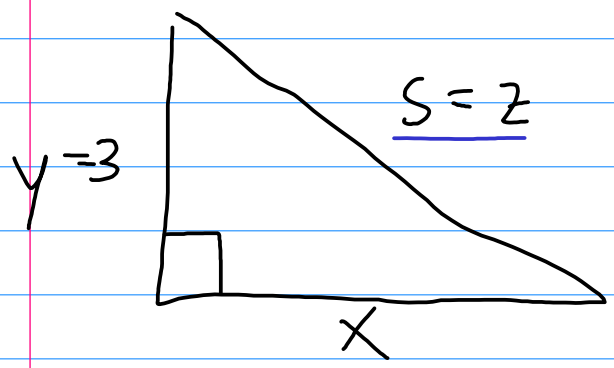
Funciones trigonometricas



$$\theta + \beta + 90 = 180$$

$$z^2 = x^2 + y^2$$

$$c^2 = a^2 + b^2$$



$$z^2 = x^2 + y^2$$

$$s^2 = x^2 + 3^2$$

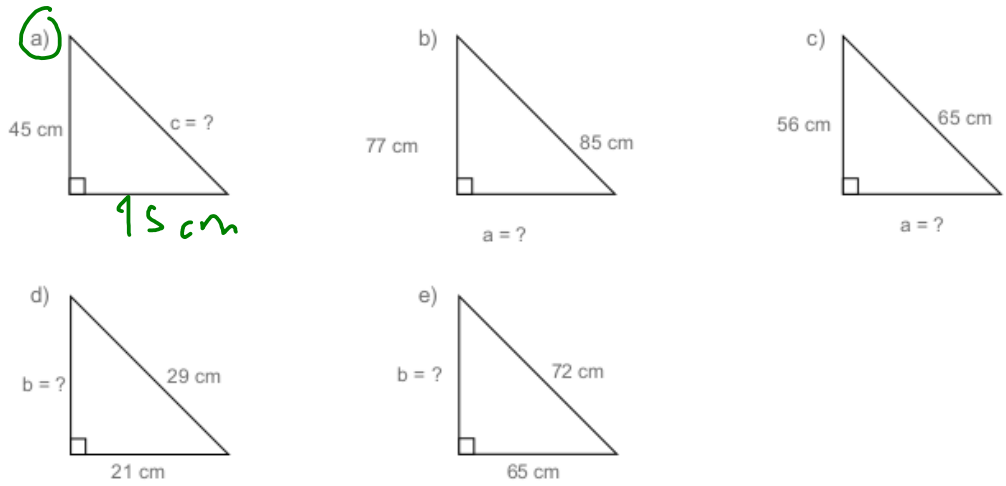
$$2s = x^2 + 9$$

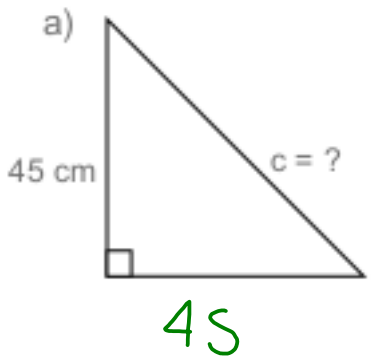
$$x^2 = 2s - 9$$

$$x^2 = 16$$

$$x = \sqrt{16}$$

$$x = 4$$





$$z^2 = x^2 + y^2$$

$$c^2 = 45^2 + 45^2$$

$$c^2 = 2025 + 2025$$

$$c^2 = 4050$$

$$c = \sqrt{4050}$$

$$= \sqrt{2025(2)}$$

$$= \sqrt{2025}\sqrt{2}$$

$$c = 45\sqrt{2}$$

$$\sqrt{50} = \sqrt{25(2)}$$

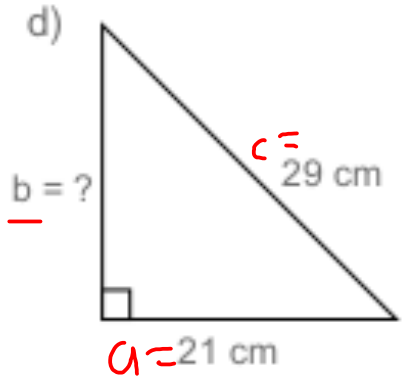
$$= \sqrt{25}\sqrt{2}$$

$$= 5\sqrt{2}$$

7.07...

7.07...

d)



$z^2 = x^2 + y^2$

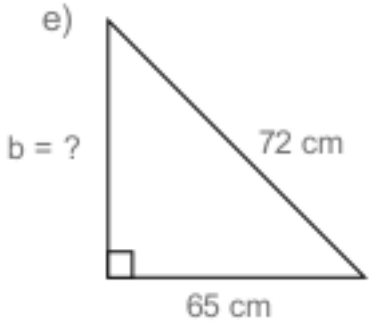
$c^2 = a^2 + b^2$

$29^2 = 21^2 + b^2$

$b^2 = 400$

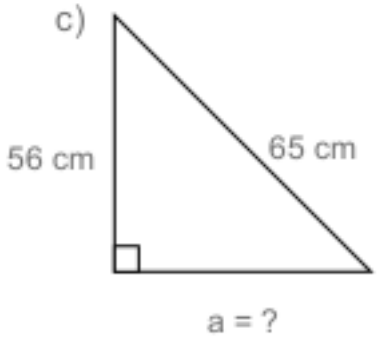
$b = 20$

e)



$b = \sqrt{959}$

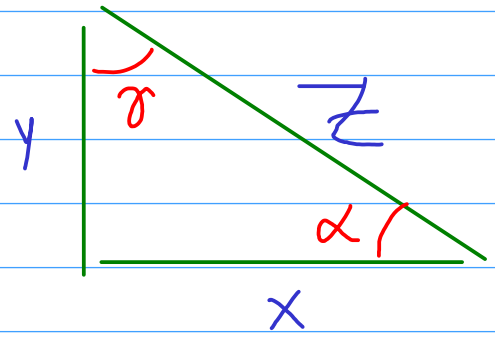
c)



$a = 33$

funciones trigonometricas

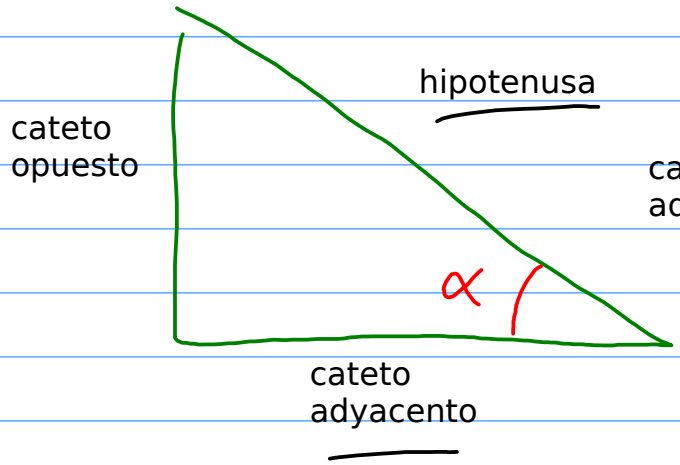
Sen(x) Cos(x) tan(x)

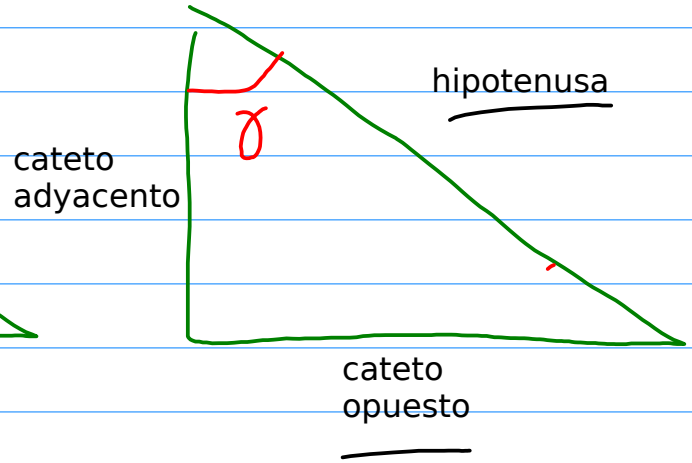


$$Sen(\alpha) = \frac{\text{cateto opuesto}}{\text{hipotenusa}}$$

$$Cos(\alpha) = \frac{\text{cateto adyacente}}{\text{hipotenusa}}$$

$$\tan(\alpha) = \frac{sen(\alpha)}{cos(\alpha)} = \frac{\frac{co}{h}}{\frac{ca}{h}} = \frac{co}{ca}$$





Sen(α) Sen(γ)

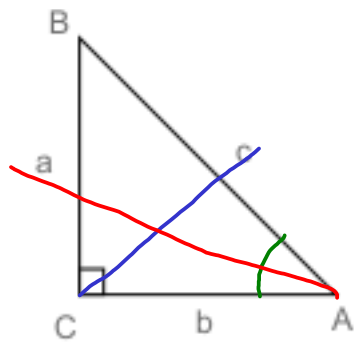
	0	30	45	60	90
sin	$\frac{\sqrt{0}}{2}$	$\frac{\sqrt{1}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{4}}{2}$
cos	$\frac{\sqrt{4}}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{1}}{2}$	$\frac{\sqrt{0}}{2}$
tan					

	0	30	45	60	90
sin	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
tan	0	$\sqrt{3}/3$	1	$\sqrt{3}$	?

$$\tan(45) = \frac{\sin(45)}{\cos(45)} = \frac{\frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{2}} = 1$$

■ Identifica la hipotenusa, el cateto adyacente y el cateto opuesto al ángulo A:

a)

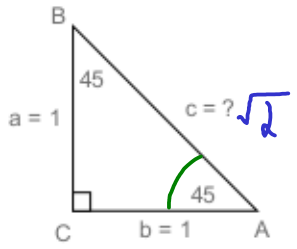


Cateto opuesto: a

Cateto adyacente: b

Hipotenusa: c

a) Encuentra las funciones trigonométricas para un ángulo de 45°



Sen 45° = $\frac{co}{h} = \frac{1}{\sqrt{2}}$

Cos 45° = $\sqrt{2}/2$

Tan 45° = $\frac{1}{1}$

Cot 45° = $1/1$

Sec 45° =

Csc 45° = $\frac{\sqrt{2}}{1}$

$$\frac{1}{\sqrt{2}}\left(\frac{\sqrt{2}}{\sqrt{2}}\right) = \frac{\sqrt{2}}{\sqrt{2(2)}} = \frac{\sqrt{2}}{\sqrt{4}} = \frac{\sqrt{2}}{2}$$

sin → cosecante → csc

cos → secante → sec

tan → cotangente → cot

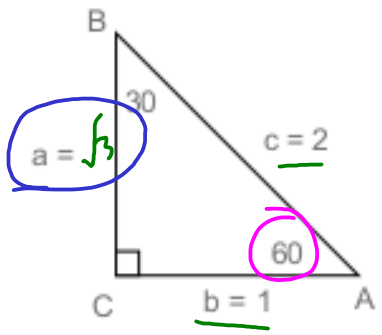
$$\csc = \frac{1}{\sin} \qquad \sec = \frac{1}{\cos} \qquad \cot = \frac{1}{\tan}$$

$$\sin(45) = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

$$\frac{2}{\sqrt{2}}\left(\frac{\sqrt{2}}{\sqrt{2}}\right) = \frac{2\sqrt{2}}{2} = \sqrt{2}$$

$$\csc(45) = \frac{\sqrt{2}}{1} = \sqrt{2}$$

b) Encuentra las funciones trigonométricas para los ángulos de 30° y 60°:



Sen 30° = $1/2$

Cos 30° = $\sqrt{3}/2$

Tan 30° = $\frac{1}{\sqrt{3}} = \sqrt{3}/3$

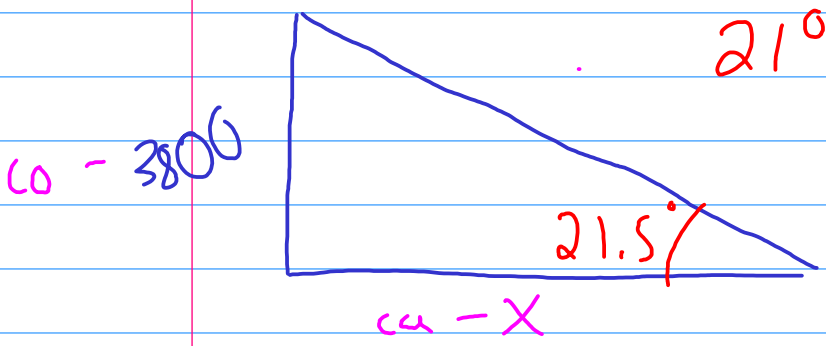
Cot 30° = $\sqrt{3}/1 = \sqrt{3}$

Sec 30° = $2/\sqrt{3} = 2\sqrt{3}/3$

Csc 30° = $2/1 = 2$

$$\frac{2}{\sqrt{3}}\left(\frac{\sqrt{3}}{\sqrt{3}}\right) = \frac{2\sqrt{3}}{3}$$

1) Un avión vuela a 3800 metros de altura sobre el nivel del mar; desde una isla se observa la nave con un ángulo de 21° 24' ¿Cuál es la distancia de la isla al avión?



$21^\circ \ \frac{24}{60} = 21.5$

$\tan(21.5) = 0.5$

2 hors 15 min = 2.25

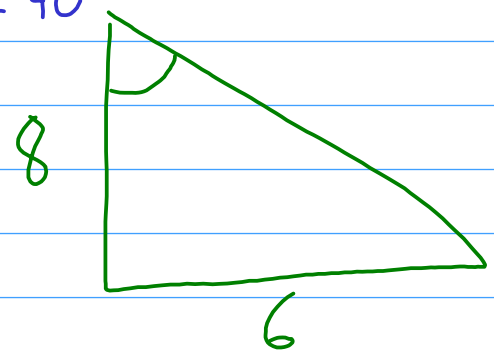
$$\tan(21.5) = \frac{3800}{x}$$

$$x \tan(21.5) = 3800$$

$$x = \frac{3800}{\tan(21.5)} = \frac{3800}{0.5} = 7600$$

2) Un mecanismo que transporta arena y tiene una altura de 8 metros llena un camión que se encuentra a 6 metros de distancia de éste. ¿Cuál es el ángulo de inclinación de la caída de la arena?

$\tan^{-1}(3/4) = 40^\circ$



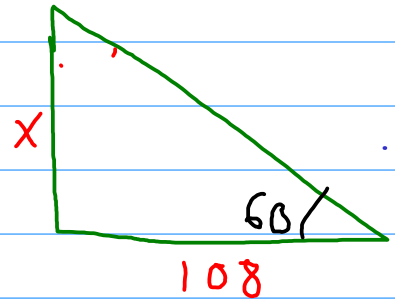
$\tan(\theta) = \frac{6}{8}$

$\boxed{\tan\theta = 3/4}$

$\theta = \tan^{-1}\left(\frac{3}{4}\right) = 40^\circ$

$\sin(30^\circ) = 1/2 \qquad \sin^{-1}\left(\frac{1}{2}\right) = 30^\circ$

4) Un edificio proyecta una sombra de 108 metros si el ángulo de elevación es de 60°, ¿cuál es la altura del edificio?



$\tan 60 = \frac{x}{108}$

$x = 108 \tan 60$

$= 108\sqrt{3}$